

The Effect of Payment Methods on Dental Service Mix: An Economic Retrospective Cohort Study

Reza Emrani¹, Katayoun Sargeran², Hossein Hessari²

¹Dental Caries Prevention Research Center, Qazvin University of Medical Sciences, School of Dentistry, Qazvin, Iran, ²Research Centre for Caries Prevention, Dentistry Research Institute, Tehran University of Medical Sciences, Tehran, Iran

Abstract

Keywords: Dentistry, fee for service, fixed payment method, health economy, payment method

BACKGROUND

The knowledge required for dentists have significantly changed. Some issues, such as demographic changes, changing patterns of diseases, new technologies, and economic considerations, can affect the combination of dental services provided.^[1] Dentists are part of the health system although much of their activities in the provision of care is driven by the “profit principle” to maintain the viability of their practices.^[2]

The supply of dental services is frequently associated primarily with the time spent by the dentist.^[3] The type and frequency of services are referred to as service mix.^[4] Factors related to the dental practitioners’ practice style and patients are associated with the service mix in dentistry. Though several studies have attempted to explain the variation in service patterns, most studies have modeled these factors separately or have included only a limited subset of potential sources of influence on service patterns.^[5] Variation in service rates is also related to the dimensions and types of practices.^[6]

Dentist-related factors include the dentist’s practice beliefs, clinical decision-making, and the dentist–patient interaction. Age and the geographic location are also associated with service patterns. Patient-level factors such as nonemergency visits, dental insurance, and socioeconomic status are related to variations in service rates.^[6–8] Economic literature offers theoretical insight into the effectiveness of various remuneration methods in influencing the dentists’ behavior and job satisfaction.^[9,10] Payment mechanisms have significant effects on clinical decision-making.^[11,12]

Payment systems have been manipulated in an attempt to achieve policy objectives and as a strategy to improve the quality of care and management, cost containment, health

Address for correspondence: Hossein Hessari, First Floor, Research Centre for Caries Prevention, Dentistry Research Institute, Qods Street, Enghelab Avenue, Tehran. 1417614411, Iran.
E-mail: h-hessari@tums.ac.ir

Received: 29 July 2021 **Revised:** 30 August 2021
Accepted: 11 November 2021 **Published:** 21 December 2021

This is an open access journal, and articles are distributed under the terms of the Creative Commons Attribution-NonCommercial-ShareAlike 4.0 License, which allows others to remix, tweak, and build upon the work non-commercially, as long as appropriate credit is given and the new creations are licensed under the identical terms.

For reprints contact: reprints@medknow.com

How to cite this article: Emrani R, Sargeran K, Hessari H. The Effect of Payment Methods on Dental Service Mix: An Economic Retrospective Cohort Study. *Dent Hypotheses* 2021;12:185-8.

Access this article online

Quick Response Code:



Website:
www.dentalthypotheses.com

DOI:
10.4103/denthyp.denthyp_101_21

indicators, and efficiency.^[13] There are two major categories of payment methods in dentistry: (1) salary or fixed payment and (2) fee for service (FFS). Primary care dentists are more likely to accept salaried employment in underserved areas compared to the FFS system, as a fixed income provides more financial security.^[14] In a salaried system, the dentist's income depends on the number of hours worked. This payment method is expected as a result to be associated with reduced activity, increased referrals to specialists and hospitals as dentists concentrate on patients who bring with them the lowest demands.^[15]

The FFS and target payments link payment to output and are based on the number of patients. FFS improves clinical activity and is associated with earlier restoration of caries, more frequent appointments, and less preventive advice given to patients. The most important risk for the patient is that the dentist may prescribe excessive treatments and interventions.^[16] Overtreatment occurs among dentists paid by FFS and under treatment may occur in salaried employment where over and under treatment are measured by comparing the prevalence of "complex" treatments with explicit guidelines for those treatments.^[17] The aim of this study was to describe the service mix of private general practitioners and to investigate the association of service mix with the method of payment in one dental clinic that had both of payment methods.

METHODS

In this retrospective cohort study, a clinician from the suburbs of Tehran with more than 35 years of work experience and a history of changing management and payment method was approached. The clinic was first established under the salary system, but the owner decided to change the payment system to the FFS and to outsource its management in 2018. This clinic was selected because it experienced a change in the payment method and had a good paper and computerized archive. The head of the clinic consented to the study. This study was approved by the Ethics Committee of the Tehran University of Medical Sciences (code 42794260).

The records of 36 dentists who worked in both periods of payment methods, nine dentists who worked only in the FFS method, and nine dentists that were only in salaried employment were reviewed. We randomly select patient records from October 1 to 30, 2018 for the salary period and March 1 to 30, 2019 for the FFS period ($n = 85$ records per dentist for each period). The total number of files was 3060.

In case of incomplete information or ambiguity, the records were excluded from the study. Data of specialized services, including orthodontic treatments, implants, and crown lengthening surgery, records of patients under 18 years old, and charity services were excluded. The number of services was then calculated including single visits without treatment, restorations, endodontic treatments, extractions, impacted tooth surgery, radiography without treatment, scaling, and patient referrals. Next, dental services were categorized into four groups:

- (1) practices without treatments (visits, diagnostic radiography, and referrals);
- (2) practices included preventive treatments (simple restorations, and scaling);
- (3) practices included aggressive treatments (randomized controlled trial, extraction, and crown); and
- (4) practices included highly aggressive treatments (surgery).

The differences in dental service mix were compared among 36 dentists. Dentists Nos 1 to 12 worked in both payment methods, dentists Nos 12 to 24 worked only in the salary period, and dentists Nos 25 to 36 worked only in the FFS period.

group.

of the NHS in England and a redistribution of care provided

are professions, markets and countervailing
Conrad P, Fremont AM, Timmermans S, eds.
Sociology 6th ed. Nashville, TN: Vanderbilt
. p. 270-89.

R. Skill-mix in preventive dental practice-will it
be future? BMC Oral Health 2015;15:S10.

, Beazoglou TJ. Economic impact of dental
tal practices. J Dent Edu 2012;76:1045-53.

lzále S, Ventelou B. Physicians' incentives to
medicine: experimental evidence. Documento
.

DS, Szuster FS, Goss AN. Service: mix of oral
eons in Australia and New Zeland. Int J Oral
22:310-13.

AJ, Roberts-Thomson KF, Brennan DS. Dental
service utilization: a 2-year follow-up study.
2.

J, Szuster F. Rates of dental service provision
and noncapital locations in Australian private
J Rural Health 1998;6:12-7.

t AC. Insurance coverage and the demand
ults for non-aged white adults. J HEcon

Opp M. Physician remuneration methods for
anada: expected outcomes and lessons learned.
2010;18:35-59.

ζ, Shamshiri AR, Hessari H. Job satisfaction
ng to workplace in Tehran. Front Dent 2021; 18.

J, Glenny AM, *et al.* The effect of different
ion on the behaviour of primary care dentists.
st Rev 2013;2013:CD009853.

ow does competition affect the clinical decision-
Ontario? 2018.

rook R, Kominski G, Albert PS, Wenger N. The
ethod on clinical decision-making: physician
enarios. Med Care 2004;42:297-302.

Brignardello-Petersen R, Pérez G. Pay-for-
iency in primary oral health care practices in
loncia Implantol Rehabil Oral 2015;8:60-6.

ζ. Payment systems and incentives in primary
ealthcare systems: Implications of recent
and Romania. in th ECPR conference,

Financial support and sponsorship

Nil.

Conflicts of interest

There are no conflicts of interest.

16. Freeman R, Carson SJ. GP payment schemes review, October 2014. GP Payment schemes review October 2014, 2014.
17. Carson SJ, Freeman R. Training and fairer payments would increase caries prevention in practice. Evid Based Dent 2015;16:6.
18. Voinea-Griffin A, Rindal DB, Fellows JL, *et al.* Pay-for-performance in dentistry: what we know. J Healthc Qual 2010;32:51-8.
19. Whittaker W, Birch S. Provider incentives and access to dental care: evaluating NHS reforms in England. Soc Sci Med 2012;75:2515-21.